

EEET202

THINES & TRANSFORMERS

MODULE: 4

Open Circuit & Short Circuit Tests

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Syllabus

ers –constructional details, principle of operation, EMF equation, convention, magnetising current, transformation ratio, phasor no load and on load, equivalent circuit, percentage and per unit regulation. Transformer losses and efficiency, condition for 7/A rating. Testing of transformers– polarity test, open circuit test, Sumpner's test – separation of losses, all day efficiency.Parallel transformers– numerical problems

Importance

Open circuit and short circuit test are performed for the determination of the transformer parameters like no-load loss, no-load current, voltage regulation, circuit constants etc.

These tests are performed without the actual loading

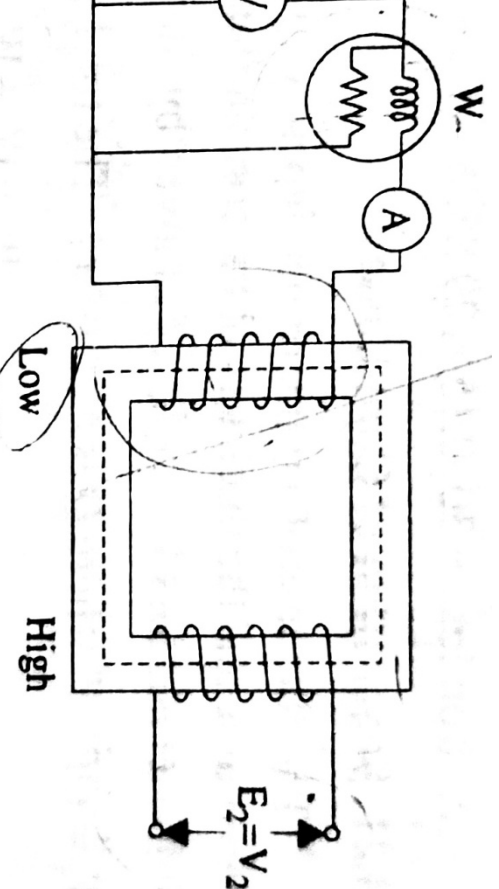
because the very less power is required for the

Circuit Test (No-Load Test)

this test is to determine the core losses and which are helpful in finding X_m and R_c . the low voltage (LV) side of the transformer. / side) is left open.

meter and an ammeter are connected in the LV figure.

applied at the low voltage side. ngs are noted.



Circuit Test (No-Load Test)

Voltage is applied at the LV side, rated flux is produced, thus core losses will occur.

Primary side is open circuited, transformer will draw no-load current, and it will be very small. (2% – 5% of rated current)

Watts will be negligibly small at LV side and nil at HV side

Power factor reading indicates the core losses of the transformer

gives no-load current.

Circuit Test (No-Load Test)

$$\text{g, } W = V_1 I_e \cos \phi_o$$

$$\frac{W}{V_1 I_e} \Rightarrow \phi_o = \cos^{-1} \left(\frac{W}{V_1 I_e} \right)$$

$$= I_e \sin \phi_o$$

$$I_e \cos \phi_o$$

$$X_m = \frac{V_1}{I_m}$$

$$R_c = \frac{V_1}{I_c}$$

Circuit Test (Impedance Test)

This test is to determine the following:

• If the transformer at any load condition
• Regulation

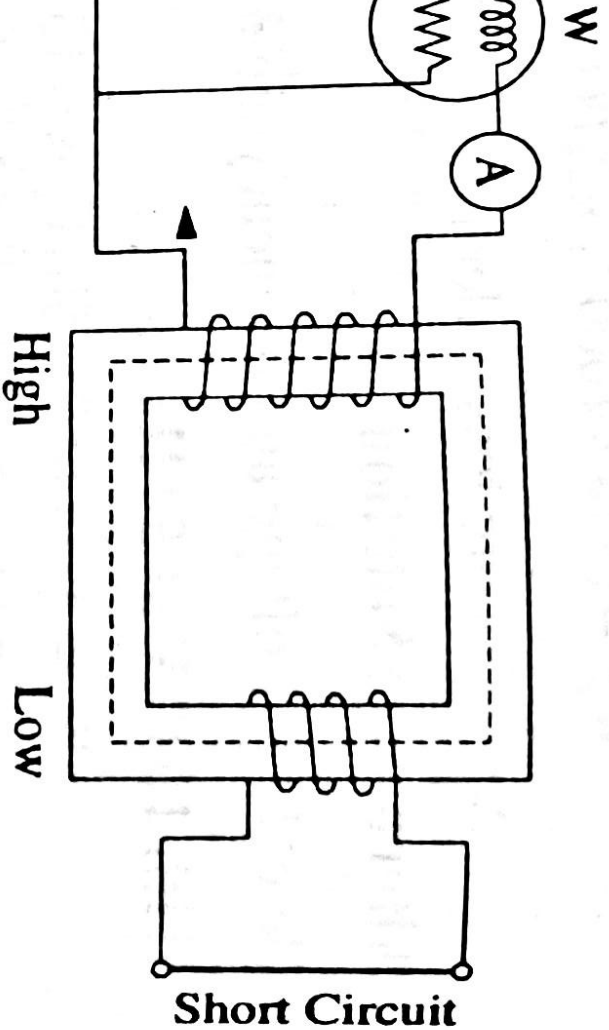
• Impedance, leakage reactance and resistance.

• The load of the transformer is short circuited by a thick

• 5% - 10% of the rated value) is applied at the HV
• The load is gradually increased till full load currents are flowing
• The following readings are noted.

• The following readings are noted.

Circuit Test (Impedance Test)



is only a small percentage of the rated value, flux is small and hence core losses can be neglected. This gives the full-load copper losses of the

Circuit Test (Impedance Test)

$$\text{g, } W = I_1^2 r_{01}$$

$$r_{01} = \frac{W}{I_1^2}$$

$$Z_{01} = \frac{V_{sc}}{I_1}$$

$$x_{01} = \sqrt{Z_{01}^2 - r_{01}^2}$$

Problem – 1

alent circuit of a 200/400 V, 50 Hz, 1-phase
the following data:

0.7 A, 70 W – on LV side

10 A, 85 W – on HV side

Problem – 1

ent circuit of a 200/400 V, 50 Hz, 1-phase
e following data:

. 7 A, 70 W – on LV side
A, 85 W – on HV side

Problem – 1

ent circuit of a 200/400 V, 50 Hz, 1-phase
e following data:

. 7 A, 70 W – on LV side
A, 85 W – on HV side